

Mark "Xavier" Bonavita

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<https://xvr6.dev/>

Education

Worcester Polytechnic Institute (WPI) | Worcester, MA

GPA 3.60 / 4.0

- **Masters of Science** in Computer Science

08/2025 - 05/2026

- **Bachelor of Science** in Computer Science

08/2022 - 10/2025

Technical Skills

Languages	C, C++, CSS, Go, HTML, Java, JavaScript/TypeScript, JSON, Python, SQL
Tools & Frameworks	AWS, Azure, Docker, Flask, Git, nginx, React, REST, Unix, Nix, SQLite, PostgreSQL, Vue

Professional Experience

AI-Powered Species Identification: Combating Illegal Wildlife Trade | Worcester, MA

Mar - Oct 2025

Lead Machine Learning Engineer & UX Consultant

- Led a National Science Foundation (NSF) funded project building an AI-driven molecular forensics pipeline that accelerates species identification for wildlife-trade enforcement.
- Achieved 95.35% classification accuracy on the target 55 species using a Vision Transformer (ViT) with transfer learning - substantially outperforming a baseline model - demonstrating the viability of machine learning for real-world conservation and anti-trafficking operations.
- Developed a cross-platform React Native app enabling real-time field identification; Deployed an efficient ViT nginx-based classification server with minimal, low-performance backend hardware that still achieved ~150ms end-to-end latency (data upload → processing → model inference → result return).
- Co-authored a research publication demonstrating how scalable machine learning pipelines can operate effectively in low-resource enforcement environments, reducing identification time from hours to sub-second responses, increasing the practicality and accessibility of molecular forensic tools.

Projects

Recruit.me, Full-Stack Job Matching & Hiring Platform | Worcester, MA

Aug - Dec 2025

- Developed a scalable full-stack job matching platform using React, AWS Lambda, and serverless MySQL, able to support thousands of concurrent users and real-time applicant/job status updates.
- Designed and implemented normalized database schemas and optimized query logic to enable fast reporting, applicant tracking, and company management, reducing backend response times by over 50%.
- Built RESTful API endpoints for job listings, applicant workflows, and admin reporting, integrating backend pagination and skill-based filtering to handle large datasets efficiently.
- Led UI/UX development with designer-provided wireframes, creating responsive interfaces for multi-role users (applicant, company, admin) and collaborating on technical documentation and UML diagrams to streamline team communication.

furspace.blog, Humanities Practicum | Worcester, MA

Jul '24 - Mar 2025

- Developed a retro-themed storytelling website inspired by early social platforms, leveraging HTML, CSS, and JavaScript to create interactive blog posts and multimedia showcases.
- Implemented custom scripts for dynamic content loading, music playback, and user-driven navigation, enhancing engagement and site usability.
- Managed deployment and site configuration on GitHub Pages, optimizing performance and accessibility while maintaining a cohesive visual identity through original artwork and iconography.

When2Goat, Meeting and Schedule Coordination Website | Worcester, MA

Mar - May 2024

- Maintained and extended a modular Flask MVC web application for event scheduling and guest management.
- Designed reusable calendar and template components, enhancing user workflow and interface consistency.
- Improved backend reliability and performance with optimized PostgreSQL queries and secure authentication via Microsoft Entra ID.
- Led feature development and bug resolution in a collaborative Agile/Scrum team environment.
- Delivered scalable, cross-device solutions supporting real-time coordination and streamlined guest access.